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ASSIGNMENT 2: BAYESIAN NETWORKS (1): SEMANTICS



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Task 1: Independence Statements

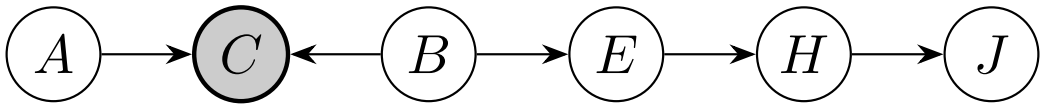
	Statement	Implied?
(a)	$I \perp I$	No
(b)	$B \perp J$	No
(c)	$F \perp G$	**Yes**
(d)	$C \perp E$	No
(e)	$A \perp H \mid J$	No
(f)	$C \perp E \mid B$	**Yes**
(g)	$G \perp F \mid I, C$	No
(h)	$J \perp A \mid C, I$	No

Only (c) and (f) follow from the graph.

“No” means independence is not guaranteed.

Task 1: Why (h) Is Not Implied

Claim: $J \perp A \mid C, I$. Observed nodes: C and I.



- Observing the collider C opens this path.
- Non-colliders B, E, H are unobserved.
- I is outside this path and cannot block it.

An active path remains: (h) is not implied.

Source: UC Berkeley CS188: D-Separation.